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IN-DEPTH RESEARCH ON USER INTERFACE AND USER EXPERIENCE ON WHAT MAKES THE DESIGN PERFECT

The engineering thesis

written under the supervision of

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Introduction

The design of websites and web applications is vital for providing an engaging and satisfactory user experience. User Interface (UI) and User Experience (UX) are essential components in the design process that significantly impact the overall satisfaction and success of the design. As digital technology usage continues to rise, understanding what makes a design perfect in terms of UI and UX becomes increasingly important.

The purpose of this thesis is to conduct in-depth research on User Interface and User Experience to identify the key elements that contribute to the perfection of design. The research will focus on both the general design of the website or web application and on the specific elements that impact the overall user experience. The study will also showcase a final example of a perfectly designed website that incorporates the key elements of UI and UX.

The objectives of this thesis are to:

1. Identify the key elements of UI and UX that contribute to the perfection of the design.
2. Analyze the relationship between UI and UX in terms of design perfection.
3. Evaluate the impact of UI and UX on user satisfaction and engagement.
4. Provide recommendations for improving UI and UX design in websites and web applications.
5. Demonstrate a final example of a perfectly designed website incorporating the key elements of UI and UX.

This study's significance lies in its contribution to understanding what makes a design perfect in terms of UI and UX. The findings of this thesis will be useful for designers, developers, and stakeholders in the website and web application industry, providing valuable insights into the importance of UI and UX in the design process.

The scope of this thesis is limited to the design elements that contribute to the user experience in websites and web applications. The technical aspects of a website or web application development will not be covered.

The structure of the thesis will consist of six chapters. Chapter 1: Problem Definition and Study Overview will provide an overview of the research topic and introduce the purpose, objectives, significance, and scope of the study. Chapter 2: Literature Review will present

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a review of previous studies on User Interface and User Experience. Chapter 3: Methodology will describe the research design and methodology used in the study. Chapter 4: User-Centred Design: Principles and Examples in Website Design will explore user-centred design principles and showcase examples of website designs that incorporate these principles. Chapter 5: Results and Discussion will present the findings and analysis of the data collected, as well as a discussion of the results. Finally, Summary will provide conclusions and recommendations for further improvements.

Chapter 1: Problem Definition and Study Overview

1.1: Background of the study

The emergence of digital technology has transformed the way we interact with websites and web applications, making them an integral part of our daily routines. This has led to an increased demand for websites and web applications that are easy to use, accessible, and aesthetically pleasing. Consequently, the importance of User Interface and User Experience design has gained immense popularity and significance in recent years.

UI design involves the design of the visual elements of a website or web application, including the layout, colour scheme, typography, and graphics. UX design, on the other hand, is concerned with how users interact with a website or web application, including ease of use, navigation, and overall satisfaction.

To ensure the success of digital products, it is essential for designers to understand the latest trends, techniques, and technologies related to UI/UX design. The design of these products should be user-centred, meaning that the design process should be focused on the needs and preferences of the users. This can only be achieved by incorporating the principles of UI design, such as consistency, simplicity, and readability, into the design process.

Despite the growing importance of UI/UX design, there has been limited research in this area. This has created a significant gap in knowledge for designers and developers, making it challenging to design digital products that are both effective and satisfying for users. Therefore, this study aims to fill this gap by providing an in-depth analysis of the key elements that contribute to the perfection of UI/UX design, which will be useful for designers, developers, and stakeholders in the website and web application industry.

1.2: Statement of the problem

As websites and web applications become increasingly integral to our daily lives, it is critical to design these products with user-centred principles that prioritize accessibility, usability, and overall satisfaction. However, despite the growing importance of User Interface and User Experience design, there remains a significant gap in knowledge for designers and developers regarding the key elements that contribute to an exceptional digital product. Although numerous studies have been conducted in this area, they often fail to provide a comprehensive understanding of the latest trends, techniques, and technologies related to UI/UX design, which is essential for ensuring the success of digital products.

This study aims to address this gap by examining a diverse range of websites and web applications across various industries to identify the critical design principles and techniques used in creating visually appealing, accessible, usable, and satisfying user-centred designs. By analysing the design, user experience, and performance of these digital products, the research seeks to provide designers and developers with a clear understanding of the critical components necessary for creating user-friendly and effective digital products.

1.3: Research objectives

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The study aims to achieve several research objectives that will contribute to a better understanding of UI and UX design principles and techniques. First, the research will identify the key elements of UI and UX that contribute to the perfection of the design. Second, the study will analyse the relationship between UI and UX in terms of design perfection. Third, the research will evaluate the impact of UI and UX on user satisfaction and engagement. Fourth, the study will provide recommendations for improving UI and UX design in websites and web applications. Finally, the research will showcase a final example of a perfectly designed website that incorporates the key elements of UI and UX. By employing a mixed-methods approach that combines qualitative and quantitative methods, including a survey, designer and user interviews, and in-depth case studies of well-designed websites and web applications, the study will provide valuable insights and real-world examples. These research objectives will contribute to the development of user-centred designs that are visually appealing, accessible, usable, and satisfying for users, and will provide designers and developers with a clear understanding of critical design components.

1.4: Scope of the study

This study is limited to the evaluation and analysis of websites and web applications in various industries to identify the critical design principles and techniques that make the design perfect. The study focuses on user-centred design, accessibility, usability, and user satisfaction as key aspects of UI and UX design. The results of this study are intended to provide guidance and insights for designers, developers, and stakeholders in the field of web design. Additionally, this study will help to identify gaps in knowledge and best practices, contributing to the ongoing dialogue around UI and UX design.

1.5: Significance of the study

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The significance of this study lies in its ability to bridge the gap between theory and practice in the field of UI and UX design. By providing a comprehensive overview of critical design elements and techniques used in creating user-centred digital products, this study will offer valuable insights and recommendations for designers and developers. The practical demonstration of a "perfectly designed website" will serve as a useful reference point for future design work. Moreover, this study will contribute to elevating the standard of UI and UX design, ultimately improving the overall user experience of websites and web applications.

Chapter 2: Literature Review

2.1: Overview of User Interface and User Experience

As a beginner researcher in the field of Human-Computer Interaction, I have come across a plethora of resources that help deepen my understanding of User Interface and User Experience design. In this subchapter, I will provide an overview of the concepts of UI and UX, and highlight some of the most valuable resources that I have found to be essential in my journey of learning about these crucial aspects of design.

UI and UX are often used interchangeably, but they are two distinct yet interrelated concepts. UI refers to the visual elements and interaction design of a product, while UX encompasses the overall experience of a user when interacting with a product, including aspects such as usability, accessibility, and emotions. Good UI design makes it easy for users to interact with a product, while good UX design ensures that the interactions are both enjoyable and satisfying.

One of the most impactful books that I have come across in my learning journey is "Don't Make Me Think" by Steve Krug (1). This book provides a clear and concise overview of the principles of UI design and how to create a user-centred design approach. Another book that I would highly recommend is "The Elements of User Experience" by Jesse

James Garrett (2). This book provides an in-depth understanding of the UX design process and how to create a successful product that meets the needs of its users.

"Seductive Interaction Design" by Stephen Anderson (3) and "About Face: The Essentials of Interaction Design" by Alan Cooper, Robert Reimann, and David Cronin (4) are also great resources for those interested in learning about UI and UX design. Both of these books provide a comprehensive understanding of the design principles and techniques used in creating engaging and effective user interfaces.

In addition to books, there are also several online courses and websites that I have found to be invaluable in my learning journey. Coursera (5) and Udemy (6) offer great courses such as "UI/UX Design Fundamentals" and "The Complete UX Design Course: Learn UX Design with Sketch," while LinkedIn Learning provides a course on "UX Design Fundamentals."

Smashing Magazine (7), A List Apart (8), and the Nielsen Norman Group (9) are all websites that provide a wealth of information and resources on UI and UX design. These websites offer articles, tutorials, and case studies on a variety of design topics, and are great for anyone looking to stay up-to-date on the latest design trends and techniques.

For those looking to connect with other designers and attend events to expand their knowledge, several conferences are dedicated to UI and UX design. UXPA (10), Interaction Design (11), and DesignCon (12) are all great options for those looking to network and learn from other designers and experts in the field.

Finally, portfolios are a great way to showcase your work and gain inspiration from others. Dribbble (13) and Behance (14) are all great platforms for designers to showcase their work and connect with other designers.

2.2: The Importance of User Interface and User Experience

In the digital age, UI and UX are essential for creating software products, apps, and websites. Designing them well is key to success. Both UI and UX impact the user's overall satisfaction with the product and their perception of the brand behind it. It is essential to

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comprehend the significance of UI and UX to craft a successful product that satisfies users' needs and stands out from its competitors.

The UI is the graphical layout of a product and includes the visual design, typography, images, and other elements that create the look and feel of the product. A well-designed UI is not only aesthetically pleasing but also makes the product easy to navigate and use. A poorly designed user interface can lead to user frustration and a decrease in engagement.

UX, conversely, is the user's overall experience with a product, encompassing both their emotional and cognitive reactions. Good UX takes into account the user's goals, motivations, and needs and incorporates design elements that help users achieve their goals in a meaningful and efficient way. A poor UX can result in user frustration and decreased engagement, as well as decreased brand loyalty.

The importance of UI and UX extends beyond the realm of digital products and into other areas of design and development. For example, in the automotive industry, good UI and UX in the design of a car's control panel and infotainment system can greatly enhance the driving experience and overall satisfaction with the car.

Another aspect of the importance of UI and UX is that it has a direct impact on the financial success of a product. Products with well-designed UI and UX tend to be more successful in the market and have a higher return on investment (ROI) compared to products with poor UI and UX. For example, in e-commerce, a well-designed website with intuitive navigation and clear calls to action can lead to increased sales and a higher conversion rate.

Understanding the role of UI and UX in the design and development process is critical to creating successful products that meet the needs of users and provide a positive user experience. References to the books, online courses, websites, conferences, and portfolios mentioned in the previous sub-chapter can provide a solid foundation for understanding the concepts of UI and UX and how they can be applied in practice.

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2.3: Key theories, models and concepts related to User Interface and User Experience

The field of UI and UX is complex and multi-disciplinary, drawing on a range of theories, models, and concepts from psychology, sociology, computer science, design, and more. In order to create effective UI and UX designs, it is important to understand these underlying principles.

One of the key theories in the field is Hick's Law (15), which states that the time it takes for a user to make a decision increase as the number of alternatives increases. This law is often used as a guide when designing interfaces, helping to ensure that users are not overwhelmed by too many options.

Another important concept is Affordance, which refers to the inherent properties of an object that determine its use. This concept is crucial in UX design, as it helps to ensure that users understand how to interact with an interface.

Gestalt psychology is another important theory in the field of UI and UX. This theory argues that people perceive objects and scenes as wholes, rather than as individual elements. This has important implications for design, as designers must ensure that their interfaces are easy to understand and navigate, rather than appearing cluttered or confusing.

The Model-View-Controller (MVC) framework (16) is another important concept in UI design. This framework separates the underlying data model of an interface from its presentation, allowing designers to create flexible and reusable code.

A core idea in UX design is User-centred Design. This process involves researching user needs and motivations and then using the gathered information to guide the design of interfaces. This process typically involves a series of iterations, with designers testing and refining their designs based on user feedback.

Finally, accessibility is a crucial aspect of UI and UX design. This involves ensuring that interfaces are usable by a wide range of users, including those with disabilities. This can involve using accessibility guidelines and technologies such as screen readers to ensure that interfaces are accessible to all users.

Comprehending these concepts is essential for creating intuitive and user-friendly interfaces. Designers must be knowledgeable about them to develop designs that satisfy user requirements. Applying these theories and principles enables designers to craft interfaces that are easy to comprehend, navigate, and use, providing users with a positive experience.

2.4: Previous studies and research on User Interface and User Experience

This subchapter offers a comprehensive overview of the evolution of UX and UI design and its effect on current practices. It traces the path that has brought us to the present state of UX and UI design, highlighting the contributions of various designers, researchers, and thought leaders. We discuss key milestones, providing insight into the transformations the field has gone through over the years.

2.4.1: The Importance of Jakob Nielsen's Heuristics in UX Design

Jakob Nielsen is a renowned usability expert and one of the key figures in the field of user interface and user experience design. He is best known for his work in developing a set of heuristics, or design principles, for creating user-friendly interfaces. These heuristics have been widely accepted as a standard in UX design, and are widely used as a reference by designers and researchers in the industry.

Nielsen's heuristics consist of 10 design principles (17) that provide a comprehensive framework for creating user-friendly interfaces. These principles include:

1. System status visibility: The system should provide users with appropriate feedback within a reasonable time frame to keep them informed of its progress.

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2. The system should use words, phrases, and concepts familiar to the user, rather than terms that are oriented to the system. Additionally, the system should follow conventions found in the real world, arranging information in a logical and natural order.
3. User control and freedom: Give users the ability to choose system functions without making mistakes. Provide an easily identifiable "emergency exit" for them to leave an unwanted state quickly. Make sure to support undo and redo operations.
4. Consistency and standards: Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform conventions.
5. Error prevention: Design your product with care to prevent errors from occurring. If errors are unavoidable, provide users with a confirmation option before they proceed.
6. Recognition rather than recall: Reduce the user's memory load by making objects, actions, and options visible. Avoid the need for users to remember details from one part of the dialogue to another. Instructions for using the system should be visible or easy to retrieve when necessary.
7. Flexibility and efficiency of use: Accelerators – invisible to the beginner – can often speed up the experience for the expert user, so that the system can meet the needs of both novice and experienced users. Enable users to customize frequent actions.
8. Aesthetic and minimalist design: Dialogs should not contain unnecessary or rarely used information. Each additional piece of information in a dialogue competes with the relevant data and reduces its visibility.
9. Help users recognize, diagnose, and recover from errors: Error messages should be expressed in plain language, without codes. They should precisely indicate the problem and constructively suggest a solution.
10. Help and documentation: Although it is better if the system can be used without any documentation, it may still be necessary to provide assistance and documentation. These should be easy to find, focus on the user's task, list step-by-step instructions, and not be excessively long.

These heuristics provide a foundation for designing interfaces that are usable, efficient, and effective, and have had a major influence on the development of UX design. Designers and researchers heavily rely on them, and their significance is unquestionable.

2.4.2: Fitts' Law: The Impact of Distance and Target Size

Fitts' Law (18) is a principle in human-computer interaction and ergonomics that provides a mathematical model for predicting the time required for a person to move to a target, typically with a pointing device such as a mouse or stylus. The law states that the time to acquire a target is proportional to the logarithm of the ratio of the distance to the target and the size of the target. The farther away the target is, the more time it takes to reach it. Conversely, the larger the target is, the less time it takes.

The mathematical expression for Fitts' Law can be written as:

$T = a + b * \log_2 (2D/W)$

Where T is the time to acquire the target, D is the distance from the starting point to the target, W is the width of the target, and a and b are constants that depend on the pointing device and task conditions.

Fitts' Law is popularly applied to the design of user interfaces for various gadgets, including desktop computers, mobile phones, and game systems. It helps designers determine the optimal size and placement of buttons, icons, and other interactive elements on a screen to ensure that they are easy to reach and use. By applying Fitts' Law, designers can improve the efficiency and speed of interactions with these elements, and reduce the frustration and error rates of users.

In conclusion, Fitts' Law is an important principle that provides a scientific foundation for designing user interfaces that are efficient and effective. Research has widely validated it over the years, making it a valuable tool for designers and researchers in the field of human-computer interaction.

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2.4.3: The Significance of Stanford Guidelines for Web Credibility in UX Design

The design of a website can play a critical role in determining its credibility, and previous studies have shown that the design of a website can greatly impact its perceived credibility. In order to address this, researchers at Stanford University published a set of guidelines known as the Stanford Guidelines for Web Credibility (19). These guidelines have since become a widely accepted standard in the field of user experience design and are often used as a reference by designers to ensure that websites are designed in a credible and trustworthy manner.

The Stanford Guidelines for Web Credibility provide research-backed advice on various aspects of website design, such as the accuracy and currency of the content, the use of established authority sources, and the provision of clear, easy-to-understand navigation. It is important to note that the guidelines emphasize the need for website content to be up-to-date and accurate, as this helps to build trust with visitors and increase the perceived credibility of the website.

The guidelines also emphasize clear and organized presentation of information. This means providing a straightforward navigation system that allows visitors to quickly find what they need. Additionally, the guidelines encourage designers to present information in a way that is easily scannable by visitors, such as using headings, bullet points, and short paragraphs, as this helps visitors quickly find and understand the information they are looking for.

2.4.4: The Impact of the Hawthorne Effect on User Interface Design

The Hawthorne Effect (20) is a phenomenon that has been widely studied in the context of user interface design and refers to the way in which people's behaviour can change as a result of being observed. This effect was first observed in the 1920s during a series of

experiments conducted at the Hawthorne Works factory, where it was found that the mere act of being observed caused workers to increase their productivity.

In the context of user interface design, the Hawthorne Effect can have a significant impact on the way users interact with a product. For example, if a user knows that they are being observed, they may be more likely to use a product in a particular way, or they may be more likely to pay attention to certain aspects of the product that they might otherwise ignore.

In order to take advantage of the Hawthorne Effect, designers must be mindful of the psychological and emotional factors that can influence a user's experience with a product. This may involve designing products that are aesthetically pleasing, that is easy to use, and that are well-suited to the user's needs and preferences. Additionally, designers may need to consider the way in which a user's perception of a product can be influenced by factors such as the user's mood, their level of motivation, and their prior experiences with similar products.

Previous studies and research on User Interface and User Experience provide a foundation for current practices in the field. They demonstrate the importance of considering various theories, models, and concepts to design effective and user-friendly interfaces.

2.5: Recommended Reading

2.5.1: The Design of Everyday Things

“The Design of Everyday Things" by Don Norman (21) is a fantastic book for anyone interested in user experience design. The book offers an extensive examination of the psychology of design and the impact it has on the user experience.

Norman argues that good design should be simple and intuitive, and that poor design can lead to frustration and confusion for users. He provides insights into how designers can

create user-friendly interfaces that are both functional and aesthetically pleasing. The book is well-written and accessible, making it easy to understand even for those who are new to the field.

I appreciate the practical approach this book takes to design. Norman provides useful examples and case studies to illustrate the concepts, making it an engaging read that brings the ideas to life.

2.5.2: The Psychology of Web Design

"The Psychology of Web Design" by Luke Wroblewski (22) is a must-read for anyone interested in user experience design, particularly in the context of web design. This book provides a comprehensive overview of how psychology impacts the user experience of websites and web applications.

According to Wroblewski, understanding the psychological forces that shape user behaviour is essential for successful and engaging web design. He delves into topics such as attention, perception, memory, motivation, and decision-making, providing practical advice on how designers can apply these principles to enhance user experience.

Wroblewski emphasizes the importance of understanding user needs and preferences to craft designs that meet their requirements and objectives. He also furnishes useful tips and strategies for conducting user research and testing, making it a valuable resource for students and experienced designers alike.

This book stands out for its simple writing style. Wroblewski clearly explains complex ideas, making it a great resource for both newbies and experienced designers alike.

2.5.3: The UX Book

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The UX Book: Process and Guidelines for Ensuring a Quality User Experience by Rex Hartson and Pardha Pyla (23) is a valuable guide to user experience design. It offers both foundational principles and practical techniques to create outstanding user experiences.

The authors present a comprehensive look at UX design, covering user research, prototyping, design, and evaluation. They also provide guidelines and best practices for each stage of the process, making it an excellent resource for both students and practising designers.

This book focuses on the significance of user-centred design. The authors emphasize understanding users' needs and objectives to craft effective and captivating designs that satisfy them. Additionally, they provide practical advice and methods for user research and testing, making it a great resource for anyone wanting to learn more about UX design. The authors also explain complex topics in an uncomplicated and understandable way, making it a great resource for both novices and experienced designers.

2.5.4: Emotional Design

Emotional Design: Why We Love (or Hate) Everyday Things by Don Norman (24) is a classic book exploring the role of emotions in design and their effect on the user experience. Norman emphasizes the critical importance of emotions in our interactions with products and provides designers with practical guidance on how to create designs that elicit positive emotions.

Norman's book covers three levels of design: visceral, behavioural, and reflective. He explains how each level contributes to our emotional response to a product and gives designers insight into how to design with emotions in mind.

The book's main strength is its focus on the user experience as a whole. Norman stresses the importance of considering the product's look, feel, and function when creating designs, making it a valuable resource for both beginners and experienced designers.

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2.5.5: Designing Interactions

"Designing Interactions" by Bill Moggridge (25) is a comprehensive guide to the design of interactive products and experiences. Moggridge offers an in-depth look at the history of interaction design, from early mechanical and electronic devices to modern digital products. He also examines the essential principles and practices of interaction design, with practical examples of how they've been applied to real-world products.

This book emphasizes understanding the user's perspective. Moggridge advocates that designers should consider the user's objectives, motivations, and behaviours when designing. He provides practical techniques to do this.

This book is its focuses on the interdisciplinary nature of interaction design. Moggridge explains how the field brings together psychology, engineering, and graphic design, and provides an extensive overview of the various skills and knowledge needed to be a successful interaction designer.

2.5.6: The Design of Future Things

Don Norman's The Design of Future Things (26) is an insightful book that examines the consequences of technology on our lives. Norman argues that technology is transforming the way we live and interact with the world and that it is essential for designers to be conscious of these changes and design products that are easy to use, accessible, and secure.

This book emphasizes the importance of incorporating the human element when designing technology. According to Norman, designers often prioritize technical aspects of products without considering how people will use them. He provides pragmatic guidance for designing user-centred products, which are tailored to satisfy users' needs instead of just technology requirements.

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Furthermore, this book pays attention to the ethical concerns related to technology design. Norman discusses the duty that designers have to make sure their products are secure, providing useful tips to address these issues.

Chapter 3: Methodology

3.1: Research design

In the previous chapter, we discussed existing studies and research related to UI and UX design, such as Jakob Nielsen's Heuristics, Fitts' Law, the Stanford Guidelines for Web Credibility, and the Hawthorne Effect. This chapter focuses on the research design for this study.

We have several research approaches available for conducting research on UI and UX design, including qualitative, quantitative, and mixed methods. We chose a qualitative research design for this study, which allows for an in-depth exploration of the elements that make a website or web application design perfect.

Qualitative research seeks to understand the meaning behind human behaviour and attitudes. This approach enables us to gather data through open-ended interviews, observations, and surveys that explore the users' experiences and their perceptions of the key elements of a perfectly designed website or web application.

3.2: Participants and sampling

The focus of this study is to identify the target population and select participants for research on UI and UX design. I took into account age, gender, occupation, and

educational level when making my decision. The inclusion criteria for participants included individuals with at least one year of experience using websites and web applications, and the exclusion criteria excluded individuals who were not fluent in English or had a visual or hearing impairment.

To select participants, I used purposive sampling, a non-probability sampling method that involves selecting individuals based on specific characteristics. For this study, the criteria focused on participants' experience and knowledge of UI and UX design. Those chosen had backgrounds as web designers, developers, or UX experts, and the ability to provide detailed and insightful information on the topic.

Purposive sampling enabled me to obtain high-quality data that was pertinent to my objectives and control the sample size, reducing the risk of bias. This method was most suitable for my study as it enabled me to acquire rich and valuable insights from the most suitable participants.

3.3: Data collection method

Data collection is essential for any research study, as it directly affects the quality and dependability of the data. For this study, purposive sampling was used to select participants, and data were collected solely through survey questionnaires.

To guarantee data validity, several methods were implemented. Firstly, data triangulation was used whereby data was obtained from multiple sources and compared to find any discrepancies or prejudices.

The data was recorded and securely stored to protect the privacy and confidentiality of the participants.

3.4: Data analysis methods

The main method employed for analysing data collected from participants was thematic analysis. This method was chosen due to its systematic approach to analysing data and identifying patterns and themes. The thematic analysis process included coding the data, recognizing themes, and then categorizing the data into these themes.

Data had to be prepared before analysis. This entailed cleaning the data and transforming it into a suitable format. The data cleaning process entailed removing any irrelevant or duplicate information and ensuring accuracy and consistency.

The thematic analysis process began after the data was prepared. The first step usually involves coding the data into relevant categories. This is done by reading through the data and highlighting relevant information. The second step involves discovering themes in the data. This is done by grouping the coded data into meaningful categories. Lastly, the data is categorized into themes that answered the research question. Fortunately, there are some applications that can be used to automate this process.

3.5: User Centred Design (UCD)

The user-centred design process is closely linked to the research objectives, data collection method, and data analysis methods discussed in previous subchapters. The research aims to identify the elements of an optimal design, making user feedback essential to the design process. Surveying users to acquire their opinions on the design and then using thematic analysis to detect patterns and themes in the feedback are the two main data collection and analysis methods. The findings from this analysis are then incorporated into the design process, making it a user-centred approach.

UCD is beneficial as it is tailored to user needs, wants, and desires. This approach involves numerous iterations of testing and refining the design with user feedback. It is fitting for this research due to its alignment with the research objectives and its comprehensive approach to designing a product that is both effective and user-friendly.

Nevertheless, there are potential limitations to this methodology, such as when the feedback of a certain user group is overly prioritized or the needs and requirements of

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other user groups are overlooked. To address these risks, a variety of user feedback was taken into account, including opinions from people of different backgrounds and experiences. This ensured that the final design was reflective of the needs of a wide range of users.

3.6: Ethical considerations

Conducting research requires taking into account ethical considerations to protect participants and maintain the integrity of the research. The ethical principles employed in this research included respect for persons, non-discrimination, informed consent, confidentiality, and privacy.

The process for obtaining informed consent included providing participants with detailed information about the study, such as its purpose, procedures, potential risks, and benefits, and giving them sufficient time to review it and ask questions. Additionally, measures were taken to ensure the participants fully understood the study and were able to agree to participate voluntarily.

Chapter 4: User-Centred Design: Principles and Examples in Website Design

4.1: Understanding Design Approaches

Before delving into the key stages of the User-Centred Design process, it is important to understand that there are various design approaches that can be taken when creating a product or service. Each of these approaches has its strengths and weaknesses, and

choosing the right approach depends on the specific project requirements and goals. In this section, we will briefly discuss four design approaches: Technology-Centred Design, Business-Centred Design, and Designer-Centred Design. As outlined earlier, the user-centred design methodology was chosen for this project mainly because it aligns with the research objectives and provides a comprehensive approach to designing a product that is both effective and user-friendly. By understanding these approaches, we can gain a better understanding of the reason why we chose User-Centred Design.

4.1.1: Technology-centred design

Technology-centred design, also known as system-centred design, is a design approach that prioritizes the technology or system over the user. In this approach, the designers focus on creating a design that works efficiently and effectively, without necessarily considering the needs and wants of the end user.

Unlike user-centred design, the technology-centred design does not involve a deep understanding of the user's needs, preferences, and behaviours. Instead, it is a more technical approach that aims to optimize the performance of the system or technology.

One of the key stages of technology-centred design is the design and development of the technology or system. This involves the creation of a system that is efficient, effective, and technically sound, with little regard for the user experience. The design team may focus on issues such as system architecture, programming languages, and technical specifications, rather than user interface design or user feedback.

Another key stage of technology-centred design is the testing and evaluation of the system or technology. In this stage, the designers may conduct technical tests and evaluations to ensure that the system or technology is functioning as intended. However, user testing and feedback may not be a primary consideration, as the focus is on the technical performance of the system rather than the user experience.

Technology-centred design can be used in various industries, including software development, product design, and engineering. It is especially useful in situations where

technology is the primary focus of the design, such as developing new software or hardware.

In software development, for example, a technology-centred design approach might involve focusing on the technical aspects of the software, such as programming languages, algorithms, and data structures. The design team would work to optimize these technical elements to ensure the software runs efficiently and meets the needs of the end users.

Similarly, in product design and engineering, a technology-centred design approach might involve focusing on the technical specifications of the product, such as its materials, manufacturing processes, and assembly methods. The design team would work to optimize these technical elements to ensure the product is functional, durable, and cost-effective.

4.1.2: Business-centred design

Business-centred design is a design approach that places the needs of the business or organization at the forefront of the design process. This approach is focused on achieving specific business goals and objectives, such as increasing revenue or reducing costs, rather than solely focusing on the needs and desires of the end users.

In a business-centred design approach, the design team works closely with the business stakeholders to identify the key business objectives and requirements. The design team then uses this information to develop a design strategy that is aligned with these objectives. This strategy may involve prioritizing certain design elements or features that are critical to achieving the desired business outcomes.

One of the main advantages of a business-centred design approach is that it allows businesses to more effectively leverage design to achieve their goals more. By focusing on the specific needs and requirements of the business, designers can create solutions that are more effective in meeting the desired outcomes.

However, this approach can also have limitations. For example, it may lead to a design solution that is less user-friendly or that does not fully address the needs of the end users. In some cases, this can result in a design solution that is less successful overall, as it may not be well-received by the target audience.

As one would expect, business-centred design is often used in organizations where the primary focus is on maximizing profits and achieving business objectives. This approach is particularly relevant in the context of website design, where businesses aim to increase website traffic, conversion rates, and sales. The main goal of this approach is to create a website that aligns with the business's overall objectives and goals.

Business-centred design is often used in industries where the primary focus is on generating revenue, such as e-commerce, banking, and finance. In these industries, the website serves as the primary point of contact between the business and its customers, making it critical for the website to be designed in a way that aligns with the business's overall objectives.

4.1.3: Designer-centred design

The designer-centred design also referred to as designer-led design, is an approach to design that puts the designer at the heart of the process. This approach is based on the assumption that the designer has the necessary expertise and creativity to come up with a design solution that meets the needs of the end user. In designer-centred design, the designer has complete control over the design process, from ideation to implementation.

The key characteristic of the designer-centred design is that the designer's preferences and ideas take priority over the needs and wants of the end user. While this approach can lead to innovative and unique designs, it can also result in designs that do not meet the needs of the user.

Designer-centred design is often used in industries such as fashion, product design, and architecture, where the designer's vision and creativity are highly valued. However, this

29:808389

approach is not well-suited for user-centred design, where the end user's needs and preferences are paramount.

One of the limitations of designer-centred design is that it can lead to designs that are not user-friendly or accessible. Without user feedback, the designer may overlook important features or functionality that are crucial for the end user. In addition, the designer's personal preferences and biases can influence the design, leading to a product that is not suitable for a wider audience.

Despite these limitations, designer-centred design can be effective in certain contexts. For example, if the design is intended for a niche market or a specific audience, the designer's personal preferences and expertise may be more important than user feedback. Additionally, designer-centred design can be useful for generating new and innovative ideas that may not have been considered in a more user-centred approach.

4.2: Importance of UCD

User-centred design has become increasingly important in the digital age, with user experience being a key factor for success. Gray (27) found that user-centred design can ensure a quality user experience by focusing on the needs, capabilities, and limitations of users. This focus allows for better usability and accessibility, which in turn leads to increased satisfaction among users. The user-centred design also takes into account the context of use, which is important for ensuring that the product is used in the way it was intended. Additionally, user-centred design can help to identify potential problems with a product before it is released, further improving user experience.

According to Pinandito (28), UCD can be used to create websites that not only meet the user’s needs and provide a satisfactory user experience but also maximize the potential of a website to achieve the desired business objectives. UCD principles can provide a more efficient design process by focusing on user needs and preferences, allowing for the development of more accurate user requirements and the creation of websites that are tailored to the user’s expectations. Furthermore, UCD can reduce the number of user complaints and inquiries due to its focus on user needs and satisfaction, as well as its

ability to identify potential usability issues before they arise. Additionally, UCD principles can provide a more cost-effective website design process, as it reduces the need to expend resources on redesigns and other costly changes due to usability issues. Therefore, it is clear that the application of UCD principles to website design can provide a number of tangible benefits.

According to D.H. Gustafson Jr. (29), UCD is a “holistic approach [that] requires that user perspectives be incorporated into all stages of the design.” This is a major challenge as it requires a multidisciplinary team of professionals with a variety of skills and knowledge. The team must be able to effectively communicate with each other in order to ensure that the user’s needs are met. This can be challenging and time-consuming, especially when the team comprises members with varied backgrounds. Furthermore, they must have the appropriate tools and resources to produce an effective design. Knowing the right tools and having access to the right resources is essential for successful UCD implementation, but can be difficult to acquire. UCD also necessitates that the team comprehend the user's needs and preferences. This can be challenging to establish, as it necessitates both research and testing. Finally, the team needs to be able to evaluate the design in order to ensure that it meets the user’s needs. Evaluation requires constant feedback from the user, which can be difficult to obtain.

UCD provides a foundation to make sure that users have a positive experience with your website, thereby allowing businesses to capitalize on the potential of their digital presence. Taking the time to understand user needs and design based on those needs is a key component to ongoing success in digital marketing. UCD is essential for businesses to understand the needs of their users, improve the user experience and gain long-term success in a competitive online market.

4.3: Consequences of Neglecting UCD

UCD has become a core part of website design. However, some websites neglect UCD principles, resulting in undesirable outcomes. These can range from a subpar user experience to decreased website traffic, impeding the site's success. Therefore, let's analyse the repercussions of disregarding UCD in website design.

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4.3.1: healthcare.gov

Healthcare.gov is a US government-run website that was launched in 2013 to provide Americans with access to affordable healthcare. The website was intended to be a one-stop shop where users could shop for health insurance and find information on healthcare plans.

However, the launch of the website was plagued with technical issues, leading to significant delays and frustration among users. The website was criticized for its poor design and user experience, which made it difficult for users to find the information they needed and complete their transactions.

Healthcare.gov has a design that is outdated and does not meet the needs of users. Its layout is cluttered and confusing, making it difficult for users to navigate and find the information they need. Furthermore, it lacks a clear hierarchy of information, making it hard for users to find what they are looking for.

Healthcare.gov also has a registration process that fails to adhere to the principles of user-centred design, as it requires users to provide an excessive amount of personal information and answer a series of security questions. This process can be lengthy and complex, making it inaccessible and unintuitive for many users.

To enhance the user experience of Healthcare.gov, several elements need to be addressed:

- Navigation: The navigation on Healthcare.gov can be confusing and overwhelming for users, particularly those who are not familiar with health insurance terminology. The site could benefit from a simplified and more intuitive navigation system that helps users find the information they need more easily.
- Language and Tone: The language and tone used on Healthcare.gov can be overly complex and technical, which may make it difficult for some users to understand. The site could benefit from a more conversational tone and simpler language that is easier for users to digest.

- Information Architecture: The site's information architecture could be improved to better reflect the needs of its users. For example, the site could use more user-centred labels for its navigation and organize content in a more intuitive way that makes it easier for users to find what they are looking for.
- Visual Design: The visual design of Healthcare.gov is somewhat outdated and could be improved to make the site more visually appealing and engaging for users. This could include a more modern colour palette, typography, and imagery.
- Mobile Optimization: While Healthcare.gov is technically responsive, the mobile experience could be improved to make it easier for users to navigate and find information on smaller screens.

4.3.2: Myspace

Myspace was once the most popular social networking platform, but it eventually lost its users to newer and more user-friendly competitors such as Facebook and Twitter. The website's downfall can be largely attributed to its neglect of user-centred design principles.

One of the most glaring issues with Myspace was its cluttered and confusing layout. The website tried to be all things to all people and offered a wide range of features, such as customizable profiles, music streaming, and messaging. However, this resulted in a cluttered and overwhelming interface that made it difficult for users to find what they were looking for.

Another problem with Myspace was its lack of consistency in design. Users could customize their profiles with their backgrounds, fonts, and colours, which often resulted in a jarring and inconsistent user experience. Additionally, the website had a confusing navigation system that made it difficult to find essential features.

In terms of UCD principles, Myspace failed to consider the needs and wants of its users. The website's design was driven by what the company thought was cool and cutting-edge, rather than what would make the user experience easier and more enjoyable.

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To improve the website, the following changes can be made:

- Navigation: The navigation on Myspace is complex, cluttered, and confusing. There are too many options, and users are not guided towards the most important or relevant features. The navigation should be simplified, streamlined, and follow the principle of progressive disclosure. Users should be presented with only the most relevant options, with secondary and tertiary options revealed only as needed.
- Layout: The layout of Myspace is busy and lacks hierarchy. There is no clear visual hierarchy, and the elements on the page are competing for attention. The layout should be redesigned to provide a clear hierarchy and visual flow. Important elements should be highlighted, and less important elements should be de-emphasized.
- Information Architecture: The information architecture of Myspace is not user-friendly. The information is scattered and difficult to find, and there is no clear sense of how different pieces of information are related. The information architecture should be redesigned to make it easy for users to find what they are looking for. This can be done through the use of clear categories, labels, and a well-organized site structure.
- Consistency: The design of Myspace is not consistent throughout the site. Different pages have different layouts, colours, and styles, which can be confusing for users. The design should be made consistent throughout the site to provide a clear and cohesive user experience.
- Mobile Optimization: Myspace is not optimized for mobile devices. The site is difficult to use on a small screen, and the layout does not adjust to fit different screen sizes. The site should be redesigned to be mobile-friendly, with a responsive layout that adjusts to fit different screen sizes.
- Accessibility: Myspace is not accessible to users with disabilities. The site does not follow accessibility guidelines, making it difficult or impossible for users with visual, auditory, or motor impairments to use the site. The site should be redesigned to be accessible, with features such as alternative text for images, captions for videos, and keyboard navigation.

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4.3.3: AOL.com

AOL.com is a website that was once a popular destination for internet users in the early 2000s. However, as time went by, the website has become outdated and its design failed to keep up with the modern trends in user experience.

One of the major issues with the website is its cluttered homepage. There is a lot of information displayed on the page, but it is presented in a way that is not user-friendly. The navigation menu is confusing, and users have a hard time finding what they are looking for. This lack of clear organization makes it difficult for users to quickly find what they need and can lead to frustration.

Moreover, the website uses outdated design elements, such as small text, which can be difficult to read, and overuse of graphics and animations, which can slow down the loading time of the website. This can lead to a poor user experience, as users are forced to wait for the website to load, which can result in them leaving the website and going to a competitor.

AOL.com also lacks mobile responsiveness. In today's world, more and more users are accessing the internet from their mobile devices, and if a website is not mobile-friendly, it can be a major turn-off for users. The website is not optimized for mobile devices, making it difficult to navigate and use on a smaller screen.

The list of what can be improved is pretty straightforward:

- Navigation: The navigation menu of the website is cluttered and does not follow the principle of simplicity. The user has to click multiple times to find the desired page, which can be frustrating and time-consuming.
- Content: The website is overloaded with content, which can be overwhelming for users. The important information is buried deep inside, making it difficult to find for users who are looking for specific information.
- User engagement: The website lacks interactivity and user engagement features. There is no feedback mechanism or platform for users to interact with each other, which can make the website feel outdated and less appealing to users.

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- Mobile responsiveness: The website is not mobile-responsive, which can make it difficult for users to access the website on their mobile devices. With the increase in the number of mobile users, having a mobile-responsive website is essential to improve user experience.
- Design and layout: The overall design and layout of the website are outdated, and the colour scheme is not visually appealing. The use of vibrant colours and modern design elements can make the website more visually appealing and engaging for users.

You may have noticed that all these websites have similar problems when it comes to their design. The most significant issue that stands out is their lack of adherence to user-centred design principles. Each website fails to prioritize the needs of its users and places an emphasis on aesthetic appeal rather than functionality.

All three websites suffer from poor navigation, making it difficult for users to find the information they need. The overuse of graphics and animations also slows down the site's loading time, creating a frustrating experience for users.

They fail to account for different user demographics. With an increasing number of users accessing the internet via mobile devices, it is crucial to ensure that websites are mobile-friendly. However, these websites do not optimize their design for mobile devices, leading to a suboptimal experience for mobile users.

Finally, there is a lack of consistency in the design of these websites. Consistency in design is essential as it helps users to understand how to interact with the website and find what they need quickly. However, these websites use different fonts, colours, and layouts, making it confusing for users to navigate.

4.3.4: Design Flaws: Examples

In addition to the analysis of the aforementioned websites, it's important to acknowledge that neglecting UCD is not limited to large corporations. As part of my own research, I

conducted a case study on two of my designs that I was intentionally making without following the UCD principles.

When I learn that a website does not follow the principles of UCD, I am reminded of the outdated web designs from the early days of the internet, during the Web 1.0 era.

The web 1.0 era, which spanned from the late 1980s to the early 2000s, was characterized by the development of the World Wide Web and the emergence of the first generation of websites. During this period, websites were primarily static and served as digital brochures, with limited functionality and interactivity. The focus was on providing information to users in a simple, easy-to-understand manner. Design elements were often limited, with simple layouts and basic typography, as the emphasis was on functionality rather than aesthetics.

In both of my examples, I placed a higher priority on functionality and ease of development, rather than giving equal weight to user experiences design considerations such as ease of navigation and aesthetics. In the first example, I created a non-UCD version of a camera company's website inspired by designs from the early 2000s, which I named Lensio. This version lacked a solid layout structure, and user-friendly features such as search, and had outdated graphics and typography.

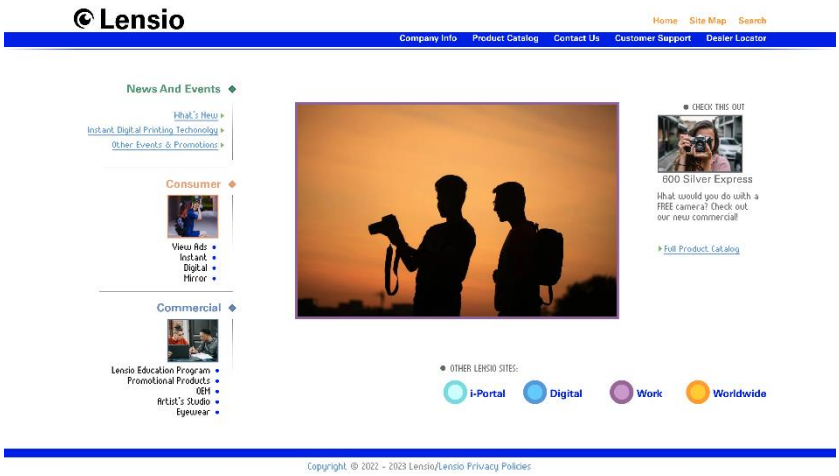

Image 1: Old Lensio Design

Source: Own Study

And for the second example, I chose an advertisement website, that was recreated after Craigslist, which is a classified ads website that has remained largely unchanged since its inception in 1995. While the website is still widely used, it has not kept up with modern

design trends and lacks many features that users have come to expect from modern websites.

I consider it to be one of the prime examples of a website that neglects UCD principles. My recreation of the website, named SellSquare, remains largely unchanged, except for a few tweaks that highlight the glaring issues with its design even more.

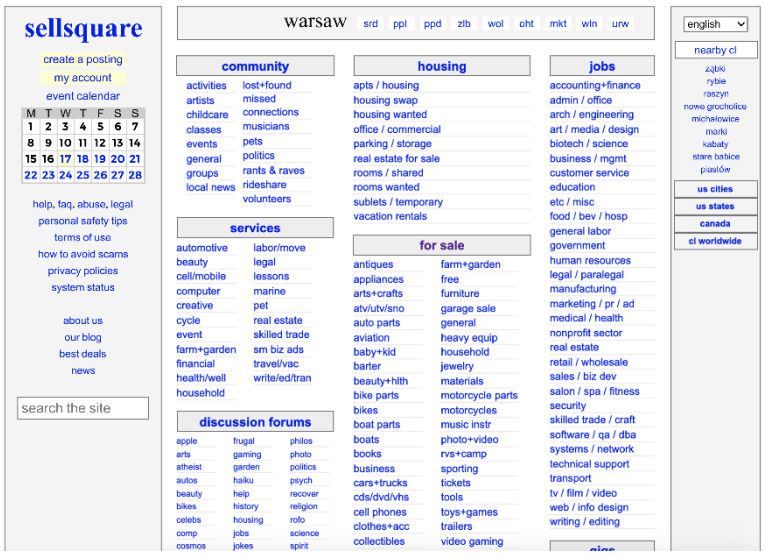

Image 2: Old SellSquare Design

Source: Own Study

The website has a cluttered homepage that can be overwhelming for new users. It lacks a clear hierarchy and has multiple links and categories that can confuse users. Moreover, the website has a dated and unappealing design, making it less attractive for users to navigate.

4.4: Examples of Successful UCD Implementation

In order to provide a solid foundation for my thesis and the survey to follow, I first decided to conduct a thorough analysis of various website elements that adhere to the principles of user-centred design. The findings were the basis for the examples I created and will help guide the evaluation of the user interface and user experience in the survey.

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4.4.1: The Impact of Icon Design on Password Fields

When designing a password field, one question that often arises is which icon to use to show or hide the password: an open eye or a slashed eye?

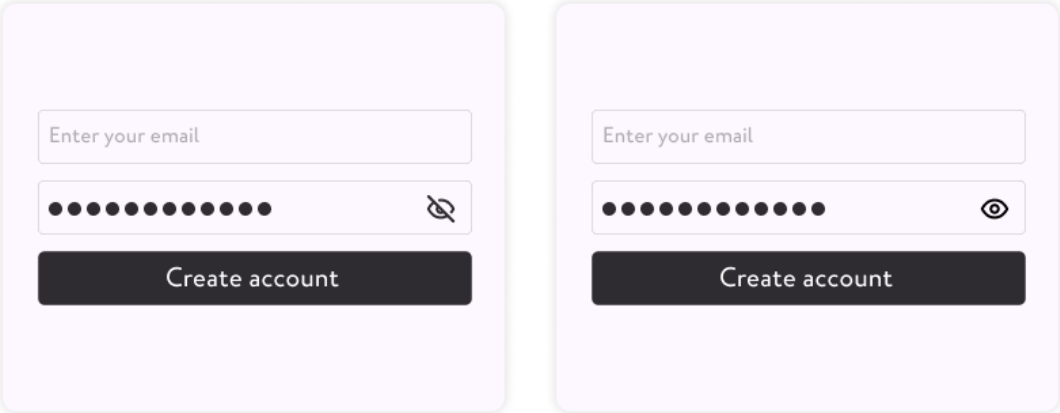

Image 3: Password Field Examples

Source: Own Study

In this example, the system provides us with two pieces of information: firstly, the visibility of the password - it is either visible or not; and secondly, the option to change the visibility setting.

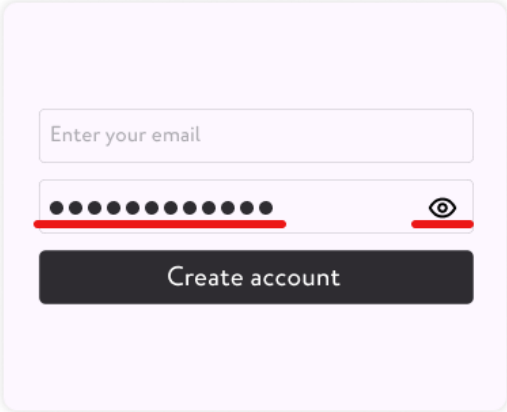

Image 4: Password Field Example 2

Source: Own Study

Hence, the icon should not indicate the current visibility status of the password since that information is already available.

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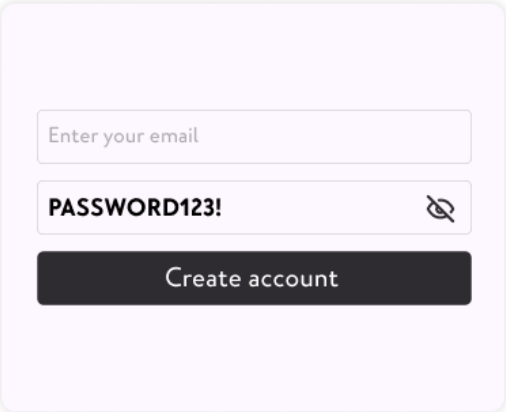

Image 5: Password Field Example 3

Source: Own Study

Rather than using the icon to indicate past actions, it is more effective to utilize the icon to convey future actions. Therefore, it is recommended to adopt an open-eye icon to represent the password's visibility status.

4.4.2: Positioning the Primary Button for Optimal User Experience

Positioning the primary button on a graphical user interface (GUI) is an important consideration for user experience. In the case of left-to-right reading languages, Windows places the primary button on the left, thereby making it the first item to be seen for left-to-right reading languages.

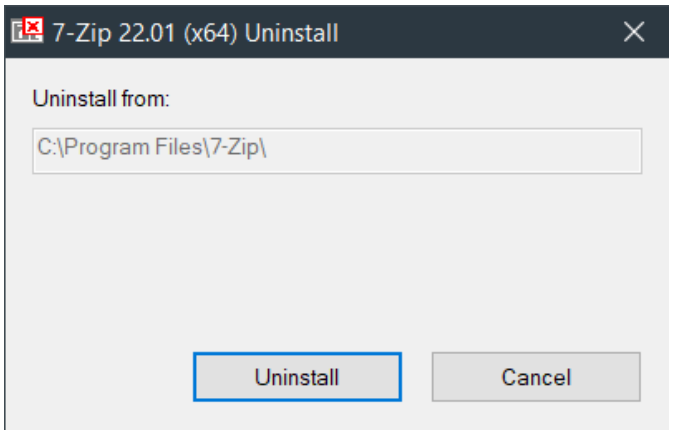

Image 6: Windows Uninstall Example

Source: Own Study

By contrast, Mac OS and Android position the primary button on the right of the GUI, which aligns with the Western convention of forward progress moving from left to right. This approach creates a sense of natural flow and may enhance the user experience for users.

Image 7: Android Settings Example

Source: Own Study

Keep it simple: move backwards with the left button and forward with the right. Place the Primary button on the right and follow the flow.

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4.4.3: The Impact of Shapes and Edges on User Perception

When designing interfaces, the choice of shapes and edges can significantly impact user perception and experience. While the use of straight edges may appear more traditional, it can trigger the Kiki effect (30).

The Kiki effect is a well-known phenomenon in psychology that was first discovered by German-American psychologist Wolfgang Köhler in 1929. It is a phenomenon that suggests that the human brain associates sharp, angular shapes with a sense of seriousness and danger, while round shapes are associated with softness and approachability.

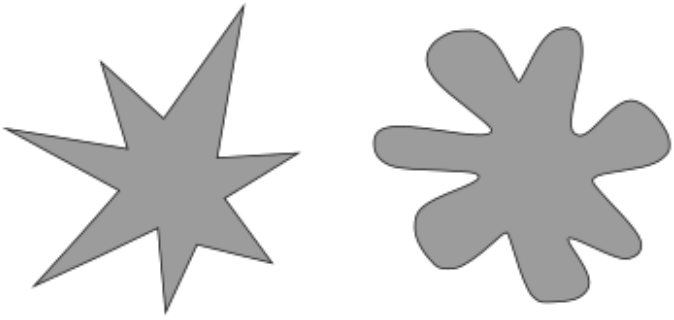

Image 8: Bouba/kiki effect
Source: https://en.wikipedia.org/wiki/Bouba/kiki_effect

In Köhler's experiment, he asked participants to match two shapes, one with a jagged edge and the other with a smooth curve, to the words "kiki" and "bouba". The vast majority of participants chose to match the sharp, angular shape with the word "kiki" and the smooth, curvy shape with the word "bouba". These results suggest that the human brain has an innate tendency to associate certain shapes with certain sounds or words, based on the sharpness or softness of the edges.

In the context of interface design, the Kiki effect can have significant implications. For example, if a button has sharp, angular edges, users may perceive it as more serious or dangerous, while rounded corners may be perceived as softer and more approachable.

Using rounded corners can help differentiate rounded buttons from the blockiness of other objects, improving usability and visual hierarchy. Rounded corners are also easier on the eyes, making it easier for users to distinguish between different elements in a design. For

example, when aligning cards in a row, it's easier to count the total number of cards when they have rounded corners.

Image 9: Corner Example 1

Source: Own Study

This is because distinguishable edges on the corners of cards guide our eyes to recognize the visual differences. Conversely, cards with sharp corners look identical and unified, which can make it more difficult for users to differentiate between them.

Image 10: Corner Example 2

Source: Own Study

In a grid layout, rounded corners perform even better. When designing a grid-based interface, rounded corners can help create a clear visual hierarchy that is easy for users to follow. The rounded corners help to guide the user's eye to each individual element within the grid, making it easier for them to locate the information they need.

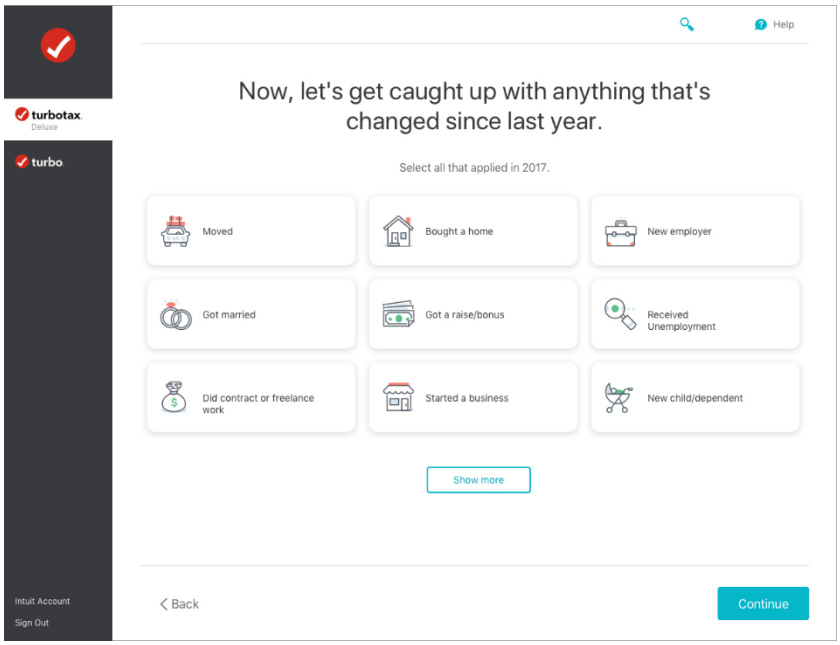

Image 11: TurboTax UI

Source: Own Study

You can see that TurboTax's Dashboard uses rounded cards that draw the eye quickly. Additionally, rounded shapes can benefit from the Contour Bias.

Contour Bias is another well-known phenomenon in psychology that suggests that people tend to prefer objects with rounded edges or contours over those with sharp edges or contours. This bias is believed to be rooted in evolutionary psychology, where humans have evolved to associate rounded shapes with safety and approachability, while sharp edges are associated with danger and potential harm.

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Image 12: Contour bias design principle by John LeMasney
Source: lemasney.com

Studies have shown that people tend to rate objects with curved edges as more visually pleasing and comfortable to hold, even if the object is not designed for comfort. For example, a study conducted by the University of Würzburg found that participants rated a chair as more comfortable and visually pleasing when its edges were rounded, compared to the same chair with sharp edges. By leveraging the Contour Bias, designers can create more engaging, user-friendly interfaces that are more visually pleasing and approachable.

While rounded shapes and edges are generally preferred in interface design, there are some situations where the use of sharp edges could be justified. For instance, this could be the case in designs for security or emergency services, where sharp edges can help convey a sense of urgency or caution.

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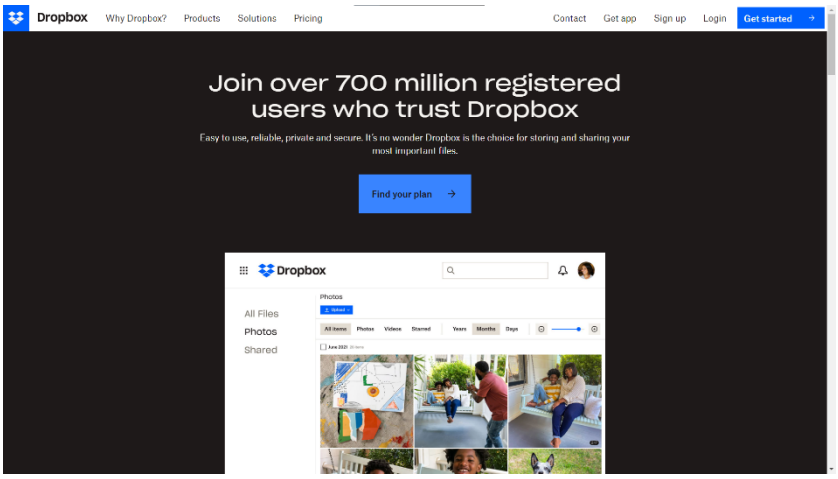

Image 13: DropBox Sharp Edges

Source: Own Study

Similarly, sharp edges can be used to create a more futuristic or modern aesthetic. This style is often associated with technology and can be found in designs for electronic devices or gadgets. Sharp edges and angles can help create a more dynamic and futuristic look, which can be visually striking and attention-grabbing.

However, even in situations where sharp edges are appropriate, designers should still consider the potential impact on user perception and experience. Sharp edges can still trigger the Kiki effect, which can be perceived as a negative or intimidating experience. Therefore, designers should always weigh the benefits and drawbacks of using sharp edges before incorporating them into their designs.

4.4.4: Search Bar Placement for Optimal User Navigation

Despite its importance, the search bar is often neglected during the website design process. Yet, it serves as a crucial tool for users to navigate the site and find the information they need. It's no secret that search functionality has become an integral part of the user experience, especially as more and more people turn to the web to find what they need. However, it's not just about having a search bar; it's about making it work well. While it may seem like a small detail, even the location of the search bar can have a big impact on usability and user experience.

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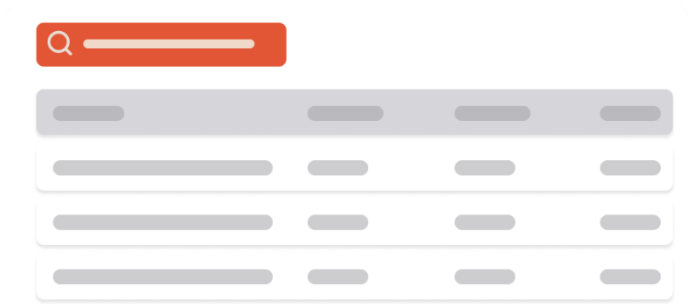

Image 14: Search Bar Example 1

Source: Own Study

Some argue that search bars on the left are quick to find, especially for users who employ an F-shaped scanning pattern when browsing a website.

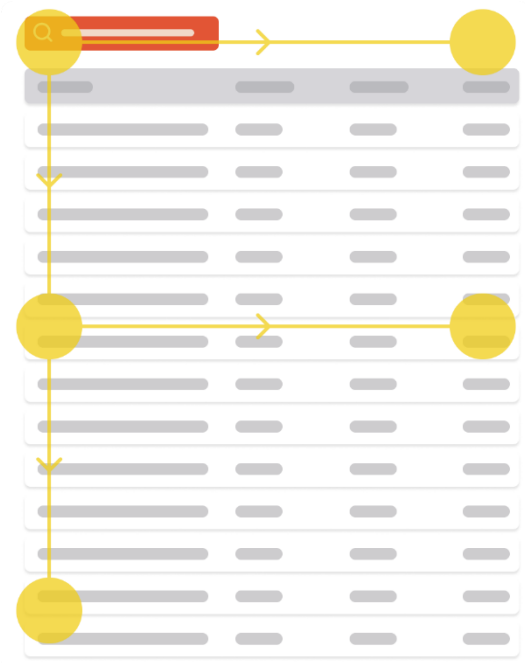

Image 15: Search Bar Example 2

Source: Own Study

Moreover, as search bars have gotten smarter, they can be good substitutes for navigation, making them a logical choice for placement on the left side of a website.

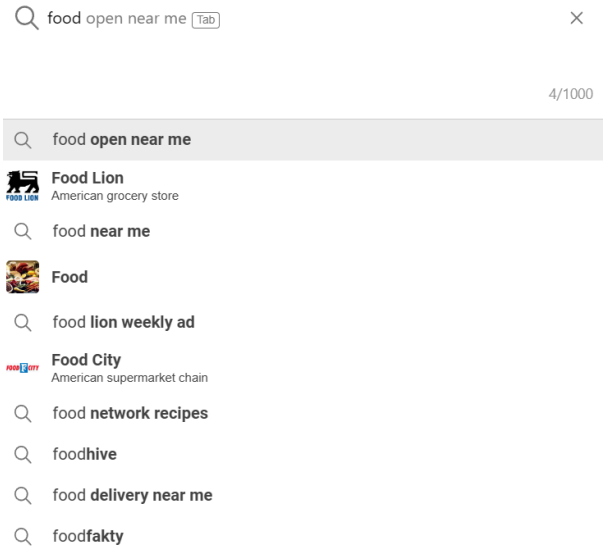

Image 16: Bing Search Example

Source: Own Study

In fact, more than half of the top 50 websites in the world put the search bar on the left, indicating that it may be the preferred location. For instance, some of the biggest names in the tech industry like Google, Baidu, Yahoo, Windows 10, and Yandex all place their search bars on the left-hand side of the screen.

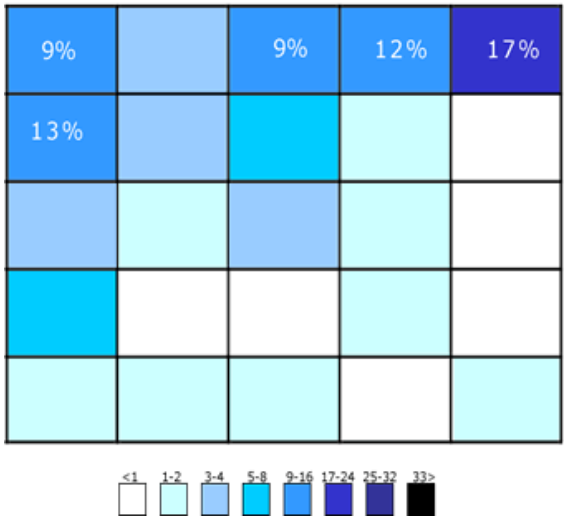

Image 17: Search Bar Expected Location

Source: https://diff.wikimedia.org/2010/06/15/usability-why-did-we-move-the-search-box/

On the other hand, a 2001 study (31) found that 29% of participants expected the search bar to be on the right, suggesting that this placement might also be valid. Placing the search bar on the right can also help prioritize more important content, drawing the user's

eye toward the centre of the page. In 2010, even Wikipedia moved its left-sided search bar to the right after its own research showed that it improved usability.

Aside from the variety of prominent websites that put their search bars on the left-hand side, there are also big names such as Dropbox, LinkedIn, Dribbble, Twitter, and Last.fm that place their search bars on the right-hand side of the screen, illustrating that the placement of the search bar is not a one-size-fits-all solution.

Noteworthy that this study is from 2001 during the early web 1.0/2.0 transition when search functionality was pretty rudimentary. The way people use and interact with search has changed drastically over the years, and it's safe to say that modern search engines have come a long way since then. While it's true that a 2001 study found that 29% of participants expected the search bar on the right, Jared Spool (32) of UIE (User Interface Engineering) argues that users are highly adaptable to finding the search bar in other locations. As he puts it, “Just because users expect the search box to be in the upper right, doesn’t mean that they can’t adjust to finding it in other places. After all, we humans are exceptionally adaptable to our surroundings.”

However, some websites or certain pages may place the search bar either right in the centre or completely to the left. While this may seem like a compromise between left and right, it's not always as straightforward. Many websites that place the search bar right in the middle heavily rely on search for their functioning, such as e-commerce sites and online marketplaces. On the other hand, some sites place the search bar right after the logo as a continuation of it, which can be a clever and intuitive placement for users. Examples of websites that place their search bars in the centre or not completely to the left include popular platforms such as YouTube, Facebook, and Reddit, as well as e-commerce giants like eBay and entertainment services such as Spotify.

In conclusion, when it comes to the placement of the search bar on a website, there is no one-size-fits-all answer. It ultimately depends on the specific needs and goals of your website. So, make your decision based on what makes the most sense for your particular context and rely on smarter searching by placing it on the left.

4.4.5: UCD Implementation in Website Design

Building on my analysis of Craigslist and Polaroid's websites, as well as my examination of successful user-centred design implementation in websites like Amazon, Airbnb, and Google, I created two website designs that prioritize user-centred design principles. These websites, Lensio and SellSquare, serve as examples of how UCD can be applied in website design to create a more intuitive and user-friendly experience.

To create Lensio, I took inspiration from the previous examples and chose to simplify the page even further. Instead of placing important content above the fold, I made while some argue that placing important content above the fold ensures that people see it right away, citing Nielsen's research from 1994 that showed only 23% of people scroll, it is also known that too much content can overload our senses. Furthermore, placing important content below the fold can make it easier to digest.

Despite the argument that forcing users to scroll can be detrimental to sign-ups, it's important to remember that quality content is key. In fact, another Nielsen research about Scrolling and Attention (33) has shown that users are more likely to scroll when presented with engaging and informative content. So, when it comes to the question of whether or not to make people scroll on a landing page, the answer is simple: let people scroll. As a result, we have achieved a balanced design that successfully follows all the principles of user-centred design while also being visually pleasing and easy on the eyes.

Image 18: Lensio New Design

Source: Own Study

For the updated SellSquare, I took inspiration from Amazon and focused on simplifying the landing page by removing unnecessary information. My goal was to create a clear and intuitive layout that guides the user to the search bar. I achieved this by yet again making the landing page scrollable to avoid cluttering the page with too much information. I also transferred the expanded categories further down the page, giving more emphasis to the search bar.

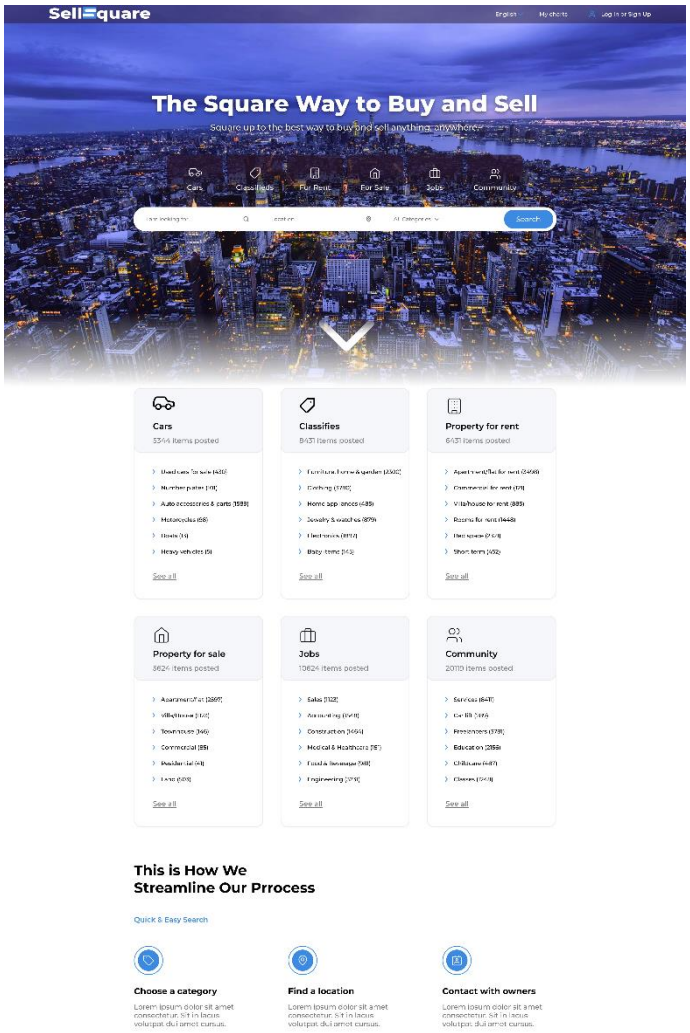

Image 19: SellSquare New Design

Source: Own Study

Chapter 5: Results and Discussion

5.1: Organizing the research

In order to conduct comprehensive research on user interface and user experience, I organized the research process into six different parts. As mentioned before, the first part was to define the research problem and the research questions that would guide the study.

Once the research questions were identified, I conducted a thorough literature review on user interface and user experience. This involved reviewing scholarly articles, research papers, and books on the topic. The aim of this review was to gain a deeper understanding of the concepts of user interface and user experience, as well as to identify any gaps in the existing literature.

After the literature review, I designed a survey using Google Forms to collect data from participants. The survey included questions about participants' age, gender, level of education, internet usage, preferred websites, and experiences with specific websites such as Polaroid and Amazon. It also included questions about the participants' opinions on two redesigned websites, Lensio and SellSquare.

To ensure the validity and reliability of the survey, I conducted a pilot test with a small group of individuals before launching the survey to a wider audience. I made necessary changes based on their feedback and redesigned questions to be clear and concise, avoiding any ambiguity or confusion. Once the survey was created, a link to the survey was generated and distributed to all participants.

5.2: Findings from the research

During the research, I gathered information from participants regarding their experience with various websites, including Polaroid, and Amazon and my designs of Lensio, and SellSquare. The participants ranged in age from 18 to 55 years old and consisted of both male and female respondents. The participants had varying levels of education, including high school diplomas, GED or high school equivalency, bachelor's degrees, and master's degrees. Some respondents were still in the process of obtaining a degree.

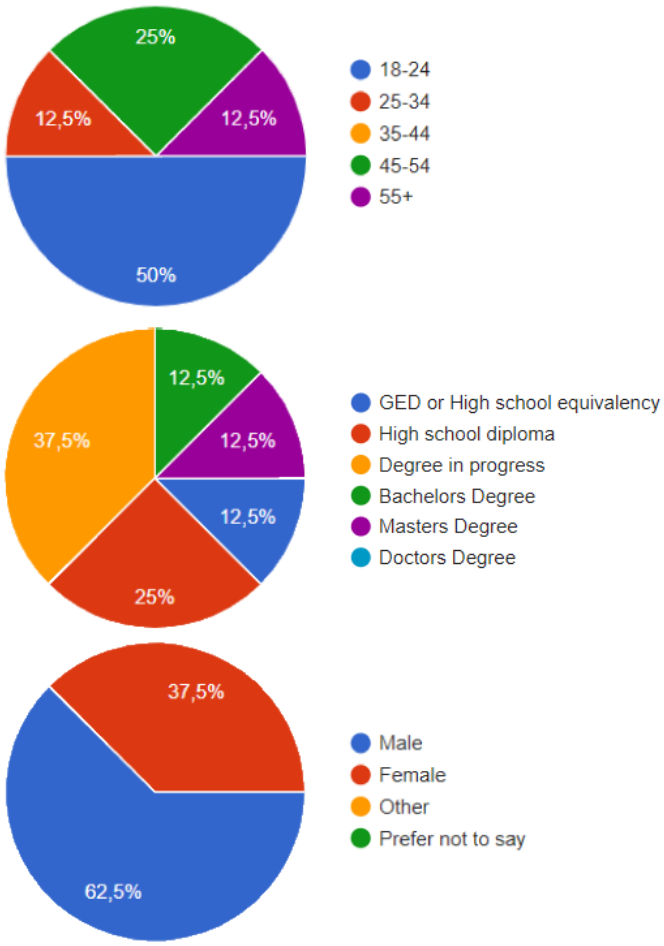

Image 20: Age, Education and Gender

Source: Own Study

The participants reported using the internet for more than 1 hour per day, with some reporting usage of more than 6 hours per day. They listed a variety of websites as their most used, including video streaming websites, social media platforms, and news websites.

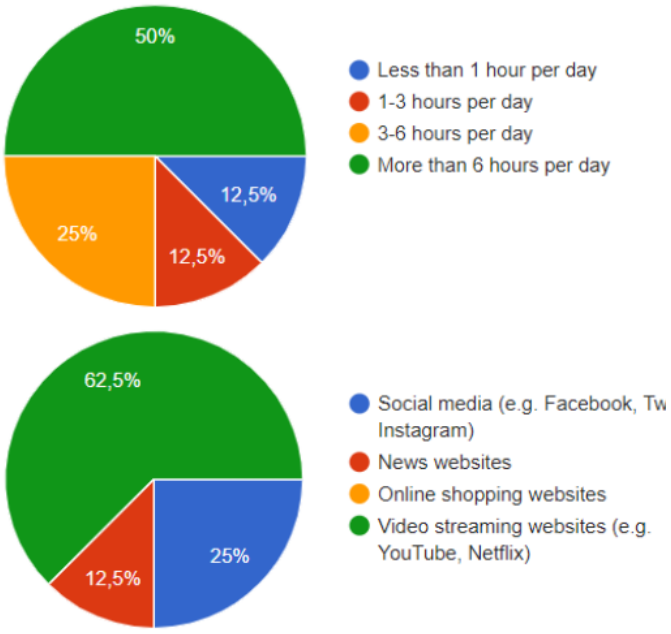

Image 21: Online Time, Main Focus of Internet Usage

Source: Own Study

Some participants reported using Polaroid or similar websites for photography equipment. The participants had varying experiences with Polaroid or similar websites, with some reporting positive experiences and others reporting difficulties with the user interface and user experience. They reported difficulties with the layout of the website and product categories.

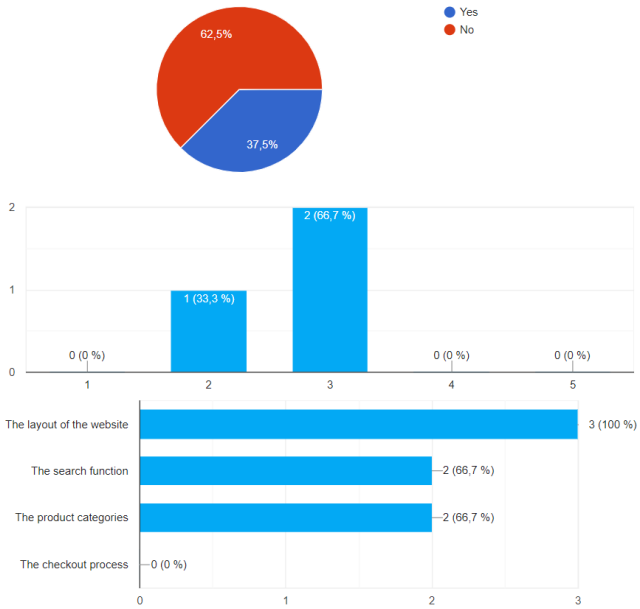

Image 22: Polaroid Usage Data

Source: Own Study

The majority of participants reported using Amazon or similar websites for buying and selling goods. They generally reported positive experiences with Amazon or similar websites in terms of user interface and user experience. However, the participants reported difficulties with the layout of the website, product categories, search functions, and checkout process.

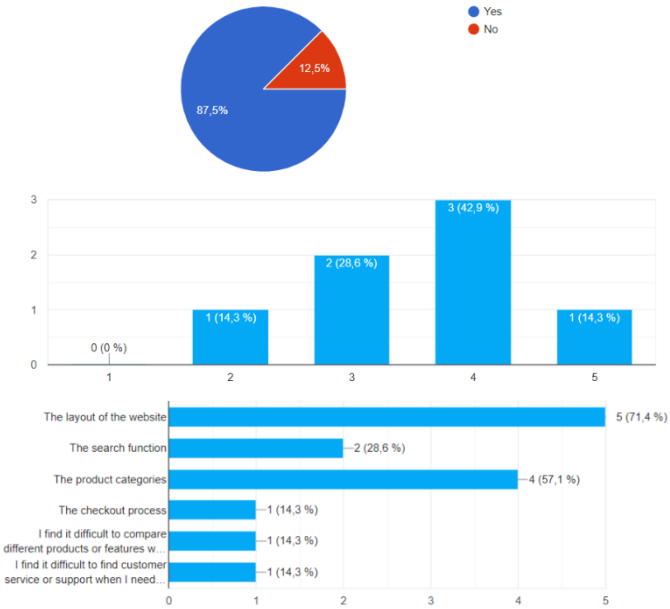

Image 23: Amazon Usage Data

Source: Own Study

Regarding Lensio and SellSquare, the participants generally liked the new design and changes made to Lensio and SellSquare. Some suggested adding more categories and making the colour palette more vibrant and eye-catching. Participants suggested adding more visual elements to the interface, but they couldn't think of anything to add to the new design for SellSquare.

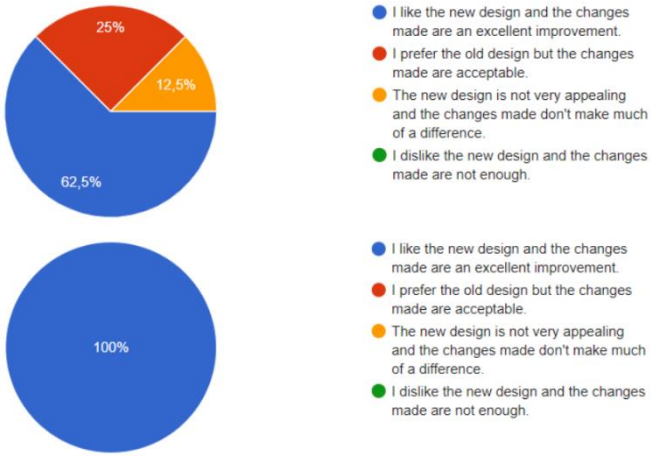

Image 24: Opinion on the new designs

Source: Own Study

Research findings reveal that user interface and user experience are crucial to the success of web pages or applications. Most participants reported positive experiences with Amazon and similar sites in terms of user interface and user experience. However, some participants cited issues with layout, product categories, search function, and checkout process.

Participants had varying experiences with Polaroid and similar sites, with some reporting positive experiences and others reporting challenges with the user interface and user experience. This emphasizes the importance of designing an intuitive and user-friendly interface, as it can significantly impact a user's overall experience with a website or application.

5.3: Comparison with previous studies

In "Research Methods in Human-Computer Interaction" edited by Jonathan Lazar, Jinjuan Heidi Feng, and Harry Hochheiser, various research methods were discussed, such as surveys, interviews, and usability testing, to name a few. These methods were used to investigate various aspects of human-computer interaction, including user experience and interface design.

Compared to this literature, my research on user interface and user experience focused specifically on website design and gathered data through surveys and interviews. While there were similarities in terms of methodology, the focus and scope of the research were different.

In terms of findings, my research revealed that user interface and user experience are critical factors in the success of websites and applications, which is consistent with the findings of previous research in the field. However, my research also highlighted the importance of specific design elements such as layout, product categories, search function, and checkout process, which may have been overlooked in previous studies.

One limitation of existing literature is that it may not fully capture the experiences and preferences of modern internet users, as technology and design trends are constantly

evolving. Further research is needed to explore the impact of emerging technologies, such as virtual and augmented reality, on user interface and user experience. Additionally, research on the impact of cultural differences on user experience can help designers create more inclusive designs that cater to diverse user groups.

Overall, my research contributes to the existing literature by providing a more focused and detailed exploration of user interface and user experience in website design. It also highlights areas where further research is needed to keep up with evolving technology and user preferences.

5.4: Conclusion on the research objectives

Despite the valuable contributions of this study, there are still a few limitations and gaps in the existing literature that should be addressed. The sample size of the participants in this study, for instance, may not be an adequate representation of the larger population, and this could affect the generalizability of the results. In addition, there is a need for further research to better understand the impact of cultural differences on user interface and user experience.

In any case, this study provides invaluable insights into the importance of user interface and user experience in the design of webpages and applications. It is hoped that the findings of this study will help designers and developers create more user-friendly and intuitive interfaces, thereby improving the user experience. Moreover, this research can also be used to inform decision-making processes related to user interface and user experience design in the development of web pages and applications.

Summary

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Throughout my research study, I aimed to investigate what makes a perfect user interface and user experience design. To achieve this, I employed a mixed-methods approach that involved collecting data through surveys, interviews, and observations.

Selecting participants using purposive sampling, I took several measures to ensure data validity, including employing data triangulation and peer review. This involved gathering data from multiple sources and comparing the results to identify any inconsistencies or biases.

During the data collection process, I implemented strict procedures to ensure accuracy and reduce bias. For instance, I held the survey in a private setting to protect the confidentiality and privacy of the participants. Additionally, I subjected my questionnaires and interview guides to a thorough review process to ensure their reliability and validity, and the data was recorded and stored securely.

My findings revealed that a perfect UI and UX design requires a balance between functionality, aesthetics, and usability. Participants emphasized the importance of clear navigation, intuitive design, and visual appeal in creating an enjoyable user experience.

I believe that my study has made a valuable contribution to the body of knowledge on UI and UX design. Going forward, future research could explore the impact of cultural differences on user preferences and the use of emerging technologies in UI and UX design.

Based on the findings of my study, I recommend several areas for improvement. Firstly, it would be beneficial to expand the sample size to include a more diverse range of participants. This would provide a more comprehensive understanding of user preferences and behaviours.

Secondly, additional methods of data collection, such as focus groups or user testing, could be incorporated to gain more in-depth insights into user experiences and perceptions.

Thirdly, emerging technologies, such as virtual or augmented reality, could be explored to enhance the user interface and overall user experience.

Lastly, collaboration with industry professionals and experts could be considered to gather their insights and perspectives on user interface and user experience design. This could lead to more innovative and effective design solutions.

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Overall, my research study has shed light on what constitutes a perfect UI and UX design, and the potential areas for improvement.

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